

Deepak Mallampati

AI ENGINEER

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Kent, OH (open to Pleasanton, CA relocation)

PROFESSIONAL SUMMARY

Applied AI Engineer with 4+ years building and shipping production **Generative AI, agentic, and LLM/NLP** systems - from a shipped consumer GenAI product to enterprise platforms serving 100k+ requests/day. Built a conductor-subagent agentic framework (42% token reduction at sustained accuracy), RAG pipelines with 92% precision and 40% hallucination reduction, and fine-tuned domain NLP models (BERT/RoBERTa, 89% F1). Fluent in Python, SQL, and cloud-native deployment (AWS/GCP/Azure, Docker, Kubernetes), turning generative-AI research into reliable, scalable product features.

SKILLS

Agentic AI Systems • LLM Applications • NLP • Generative AI • RAG Architecture • Production ML Deployment
Generative AI & LLMs: GPT-4o, GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | Agentic & Multi-Agent Systems | RAG Architecture | Prompt Engineering | Fine-tuning (LoRA, QLoRA, PEFT) | Hugging Face Transformers
Machine Learning & NLP: PyTorch, TensorFlow, Keras | NER, Text Classification, Summarization | Deep Learning | Model Optimization | Drift Detection | RAGAS, BERTScore evaluation
Vector Search & Retrieval: FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Semantic Caching
Cloud & MLOps: AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure (OpenAI, AI Services), GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD | MLflow, Weights & Biases | Monitoring & Observability
Data & Languages: Python, SQL, JavaScript/TypeScript, Java | PostgreSQL, MongoDB, Redis | Spark, Databricks, Kafka | Pandas, NumPy | Data Preprocessing & Feature Engineering | FastAPI, React | Agile/Scrum

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer

USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to benchmark model iterations before deployment and surface regressions before release.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Developed **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across **25+ AI agents** and workflows.
- Built **12 APIs and microservices** powering new generative-AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing platform model-selection decisions.
- Engineered content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining response quality.
- Partnered with product, platform, and data engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 using a curated instruction dataset, conditioning the model on a six-stage narrative schema for controllable long-form generation.
- Built a **privacy-preserving ML pipeline** on a clinical dataset (k-anonymity, l-diversity, differential privacy), cutting re-identification risk from 3.38% to 0.32% while retaining 99% of baseline predictive performance.

Emerson

Jun 2022 - Apr 2023

ML Trainee

India

- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification on domain-specific datasets, achieving **89% F1-score** on custom NER tasks.
- Developed regression and classification models on operational data; engineered features and executed end-to-end **data preprocessing** to produce model-ready training pipelines.

PROJECTS

MangaLens Shipped

A live Chrome extension delivering real-time **generative-AI translation** of manga and webtoons across 29+ sites - turning a multi-step manual chore into a single inline action.

LLM translation · Chrome Web Store · real-time inference

career-ops

An autonomous, **agentic** AI job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision.

Agentic orchestration · Claude Code · automation

Story Director LLM

A fine-tuned language model that generates structured short-form video narratives from a single prompt, paired with an automated rendering pipeline.

QLoRA fine-tuning · generative pipeline

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025

GPA: 3.70 / 4.0